MIL-H-865D
13 December 1976
SUPERSEDING
MIL-H-865C
9 July 1964
(See 6.3)

MILITARY SPECIFICATION

HOLDERS, ELECTRODE, SHIELDED METAL ARC WELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers welding electrode holders for use in the shielded metal arc welding process.

1.2 Classification. Electrode holders shall be of the following types, sizes, and classes, as specified, (see 6.1):

Type I - Positive grip.

Size 1 - 300 amperes, minimum.

Class 1 - Insulated, fixed.
Class 2 - Semi-insulated, fixed.
Class 3 - Semi-insulated, detachable.

Size 2 - 500 amperes, minimum.

Class 1 - Insulated, fixed.
Class 2 - Semi-insulated, fixed.
Class 3 - Semi-insulated, detachable.

Type II - Clamp, hinged, spring loaded.

Size 1 - 300 amperes, minimum.
Size 2 - 500 amperes, minimum.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-W-45562 - Welding and Soldering Equipment Supplies and Accessories, Packaging of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, D. C. 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3439

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3. REQUIREMENTS

3.1 The materials used in the construction of all components shall be capable of completing the examinations and tests specified herein. All components shall be new and unused. Repaired or reconditioned components shall not be acceptable.

3.2 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of holders which are otherwise in accordance with this specification.

3.3 Design. Design of electrode holders shall be such that no arcing of the gripping surfaces will occur when electrode is in contact with material being welded, or when carrying current specified in table I. Ejection of electrode stub shall be such as to be accomplished without excessive manipulation.

TABLE I. Physical and performance characteristics.

Type	Size	Rating (minimum) (Amperes)	Range of electrodes (Inch)	Class	Weight (including cable connection) (ounces)
I	1	300	3/32 - 1/4	1 and 2 3	23 28
I	2	500	1/8 - 5/16	1 and 2 3	23 28
II	1	300	3/32 - 1/4	--	23
II	2	500	1/8 - 5/16	--	34

3.4 Interchangeability. The parts of an electrode holder shall be interchangeable with similar parts of other holders of the same type and brand. Assembly and replacement of parts shall be accomplished without the use of special tools.

3.5 Current leakage. Maximum leakage permissible shall be 5 milliamperes when tested in accordance with 4.5.2 and 4.5.3.

3.6 Heating. Temperature rise of the holder shall not exceed 75°F (24°C) above ambient temperature when tested in accordance with 4.5.5.

3.7 Water absorption. Holders shall not swell, crack, warp, or show undesirable characteristics to the extent that binding occurs or their serviceability is adversely affected, when tested in accordance with 4.5.2 and 4.5.3.

3.8 Impact test. No part of the electrode holder shall break, crack, show evidence of a tendency to spall-off, or become unserviceable after being subjected to the test specified in 4.5.4.

3.9 Connections. Electrode holders shall be designed to permit connection of the welding cable to the holder by mechanical means or by brazing with a silver base alloy or thermit welding.

3.10 Marking. Each holder shall be marked in a permanent and legible manner with the manufacturers name or trademark so that the source of manufacture may be readily determined. In addition, the holders shall be marked with the manufacture's type/model number and the electrical rating.

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3.11 Type I, positive grip. Type I holders shall be similar to figure 1 and meet the requirements of table I.

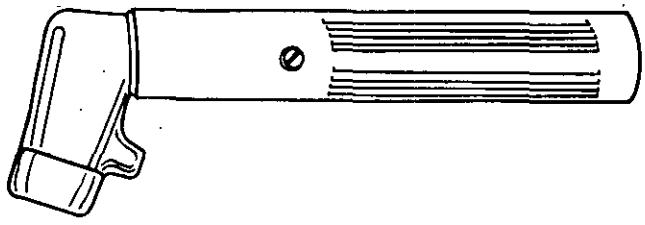


FIGURE 1. Type I, positive grip holders

3.11.1 Class 1 - Insulated, fixed. Both the handle and the electrode grip-end of the holders shall be fully insulated; no uncovered countersunk, or exposed screws or similar material shall be permitted on any part of the electrode holder. Material used for covering such exposed parts shall withstand the examination of 4.4 and the tests specified in 4.5.1 through 4.5.6. The holders shall be capable of holding the electrode in a position within nondetachable 109 to 131 degrees to the axis of the holder. Electrode grip-end shall be from body of holder.

3.11.2 Class 2 - Semi-insulated, fixed. Holders shall be insulated so that contact with a plane surface, such as ship's structure, cannot establish an electrical circuit. Heads of screws used for attachment of grip-end insulation shall be countersunk at least 1/8 inch below the surface in holes not exceeding 7/16 inch diameter. Perforations for ventilation or holes for access to screws are permitted in the handle provided cable exposed thereby is insulated at least 3/8 inch below the handle. In addition, the size and shape of the perforations or holes in the handle shall be such that contact of exposed cable with the hand shall not be permitted. Holders shall be capable of gripping electrodes in the following positions with respect to the axis: 90 degrees, 109 to 133 degrees, and 180 degrees. Electrode grip-end shall be nondetachable from body of holder.

3.11.3 Class 3 - Semi-insulated, detachable. Class 3 holders shall conform to class 2, except that holders shall be so constructed that the electrode clamping device may be quickly detached from or securely attached to the body of electrode holder without the aid of tools.

3.12 Type II, clamp hinged, spring loaded. Type II holder shall be similar to figure 2 and meet requirements of table I.

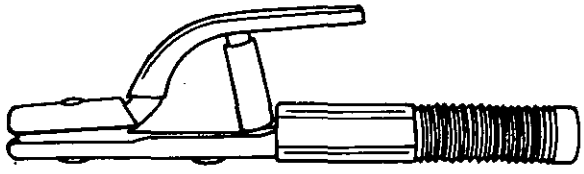


FIGURE 2. Type II, clamp hinged, spring loaded holder

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3.12.1 Type II holders shall be fully insulated, with the exception of the (tong) jaws contact points with electrode. Jaws shall be of a serrated or grooved design so as to firmly hold electrodes specified in table I at a minimum of seven positions.

3.13 Workmanship. Workmanship of the holders shall be such that they shall be free of defects that would adversely affect the operation, efficiency, servicability and life expectancy of the tool.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance.

4.2.1 Lot. Electrode holders of the same type, size and class offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.2.2 Sampling. A random sample of holders shall be selected for inspection from each lot in accordance with table II, with acceptance conforming to requirements of MIL-STD-105 with an acceptance quality level (AQL) of AQL=1.5 percent defective.

TABLE II. Lot sampling.

Lot size, number of holders	Inspection level
40 and under	III
41 to 300	II
301 and over	I

4.3 Quality conformance inspection. Quality conformance inspection shall be performed on each sample holder selected in accordance with 4.2.2 and subjected to examination specified in 4.4 and the tests specified in 4.5.1 through 4.5.6. Failure to conform to the requirement of any examination or test shall be cause for rejection. If the number of such nonconforming holders in any sample exceeds the acceptance number for the samples the lot represented shall not be offered for delivery.

4.4 Examination. Each sample holder selected in accordance with 4.2.2 shall be visually examined for workmanship, design construction, materials and components. Holders shall be inspected for their capability of gripping electrodes at the positions specified in 3.11.1 thru 3.12.1. Electrodes of the smallest and largest sizes specified, shall be tested in the holder to determine conformance with table I.

4.5 Test procedures.

4.5.1 Weight. Weight of one of the sample holders selected in accordance with 4.2.2 shall be determined by weighing both the holder and a length of welding cable of the gage specified in 4.5.5 then connecting the holder to the cable and weighing the two together. The increase in weight of the connected holder and cable together over that of the cable alone shall be taken as the weight of the holder. The balance of the units in the sample without being connected to the cable may be checked against the first, making due allowance for the weight increase, if any, occasioned by the connection.

4.5.2 Water absorption test I. Each sample holder selected in accordance with 4.2.2 shall be completely immersed in fresh water for a period of 16 hours, followed by drying 3 hours at a temperature not less than 175°F (80°C) nor greater than 225°F (108°C). After drying, a current leakage test shall be performed in accordance with 4.5.2.1 and followed by tests in 4.5.3. Any evidence of swelling, cracking, warping or undesirable characteristics shall be cause for rejection in accordance with the requirements of 3.7.

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4.5.2.1 Current leakage. Each holder and grip shall be wrapped with aluminum foil so that they are connected and subjected to a potential of 1200 volts alternating current (r.m.s.), 60 Hertz (Hz), applied between conductive inner parts and the foil for a period of one minute. Measure the current leakage between the conductive inner parts of the holder and the foil. Current leakage shall be no greater than 5 milliamperes as specified in 3.5.

4.5.3 Water absorption test II. Each holder tested in accordance with 4.5.2 shall be immersed in fresh water for a period of 1 hour then dried with a clear cloth and the current leakage test of 4.5.2.1 shall be repeated.

4.5.4 Impact test. Each sample holder tested in accordance with 4.2.2 shall be dropped twice on the electrode grip-end from a distance of 6 feet onto a steel plate having a minimum thickness of 1/2 inch. In addition, the holders shall also be dropped on the plate once on each side. The point of impact on the side shall be on the insulating material at a random location. The holder shall be examined for compliance with 3.8.

4.5.5 Heating. Each sample holder selected in accordance with 4.2.2 shall be attached to a welding cable of the gage specified in table III, and shall grip a solid copper wire approximating in diameter the largest size electrode for which the holder is designed. A calibrated bare thermocouple of adequate range shall be laid on the handle directly over the point of cable attachment, and shall be securely taped in position. The temperature indicated by the thermocouple shall then be raised to a value 75°F (24°C) above ambient, by passage of sufficient electric current for a sufficient length of time. After allowing a 20 second interval (approximately the time required to change electrodes), the holder shall be subjected five times to the current indicated for the periods specified in table III with a 20 second interval between termination of one application of current and the beginning of the next. The temperature 20 seconds after termination of the final application shall not exceed that immediately prior to the first of the five.

TABLE III. Heating test conditions.

Current (amperes)	Period of application (seconds)	Cable size	
		Number	Circular mil (c.n.)
300	120	1/0	105,500
500	150	2/0	133,100

4.5.5.1 Electrode Burning. Each sample holder tested in accordance with 4.5.5 shall be tested by burning continuously 13 electrodes each of the maximum size specified in table I on straight polarity and 13 on reverse polarity. Stub lengths shall be no longer than 1-1/4 inches. Holders shall show no signs of physical breakdown or overheating as specified in 3.6.

4.5.6 Inspection of the preparation for delivery. Sample packages, packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)

5.1 Preservation-packaging, packing and marking. Electrode holder shall be individually preserved-packaged level A or C, packed level A, B, or C as specified (see 6.1) and marked in accordance with MIL-W-45562.

5.1.1 Intermediate packaging. When level A preservation-packaging is specified a quantity as specified (see 6.1) of packaged holders of the same type, size and class shall be intermediate packaged in accordance with MIL-H-45562.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size or class of electrode holder required (see 1.2).
(c) Levels of preservation and packing required (see 5.1).
(d) Quantity of packaged holders to be intermediate packaged (see 5.1.1).

6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 Changes from previous issue. The extent of changes (additions, modification, corrections, deletions) preclude the annotation of the individual changes from the previous issue of this document.

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